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Inventor(s)	Sakai; Mizuo et al.

Heat exchange unit

Abstract

An aspect of a heat exchange unit of a refrigeration cycle device, includes a housing having openings, and gripping members attached to the openings. Each gripping member has a main body having an inner space that opens to an outside, and a plurality of claws that protrude from the main body and that hook to a circumferential edge of the openings. The plurality of claws include a plurality of first claw portions, and at least one second claw portion that is provided on an opposite side to the plurality of first claw portions in a predetermined direction that intersects the opening direction. At least one claw portion out of the plurality of first claw portions and the at least one second claw portion is an elastic claw that is cantilevered by the main body and is elastically deformable in the predetermined direction with respect to the main body.

Inventors:	Sakai; Mizuo (Tokyo, JP), Noguchi; Koji (Tokyo, JP), Takahashi; Yusuke (Tokyo, JP), Kato; Nobukazu (Tokyo, JP), Tsumura; Shoko (Tokyo, JP), Murakami; Hirokazu (Tokyo, JP), Otsuka; Motoki (Tokyo, JP), Sakabe; Yudai (Tokyo, JP)
Applicant:	Mitsubishi Electric Corporation (Tokyo, JP)
Family ID:	1000008778362
Assignee:	Mitsubishi Electric Corporation (Tokyo, JP)
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Primary Examiner: Lane; Devon

Attorney, Agent or Firm: Posz Law Group, PLC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This application is a U.S. national stage application of International Application No. PCT/JP2021/028552, filed on Aug. 2, 2021, the contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a heat exchange unit.

BACKGROUND

(3) For example, a grip that is attached to a cabinet of an outdoor unit of an air conditioner that is used during transport or the like is described in Patent Document 1. A method of attaching the grip in Patent Document 1 is done by pressing and fixing grips to side plates of the cabinet by providing open recesses, into which the grips are fit, on a front surface plate and a rear surface plate of the cabinet.

PATENT DOCUMENT

(4) Patent Document 1: Japanese Unexamined Patent Application, First Publication No. H04-359722

(5) In a case similar to the case of the method of attaching a grip as mentioned above, there is a need to sandwich the grip between differing plate members of the cabinet, and the work efficiency of attaching the grip is bad. On the other hand, if openings are provided in the cabinet, and a method of inserting the grip into the openings is adopted, the work efficiency of attaching the grip improves. In such a case however, a problem of the attachment strength of the grips easily declining exists.

SUMMARY

(6) The present disclosure has been made in order to address the problem above, and an object is to provide a heat exchange unit that has a construction where it is possible to ensure an attachment strength of gripping members, while improving a work efficiency of attaching the gripping members.

(7) An embodiment of a heat exchange unit according to the present disclosure is a heat exchange unit of a refrigeration cycle device that includes a housing having openings, and gripping members attached to the openings. Each gripping member of the gripping members has a main body having an inner space that opens to an outside of the housing, and a plurality of claws that protrude from the main body, and that hook to a circumferential edge of the openings from an inside of the housing. The plurality of claws include a plurality of first claw portions, and at least one second claw portion that is provided on an opposite side to the plurality of first claw portions in a predetermined direction that intersects the opening direction of an opening. At least one claw portion out of the plurality of first claw portions and the at least one second claw portion is an elastic claw that is cantilevered by the main body, and the elastic claw is elastically deformable in the predetermined direction with respect to the main body. The plurality of first claw portions include at least one fixed claw, and at least one elastic claw, and the protrusion height of the at least one fixed claw with respect to the main body is less than the protrusion height of the at least one elastic claw with respect to the main body.

(8) According to the present disclosure, it is possible to ensure an attachment strength of the gripping members, while improving a work efficiency of attaching the gripping members.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 A schematic block diagram that schematically shows a refrigeration cycle device that is included in an outdoor unit in a first embodiment.

(2) FIG. 2 A perspective view that shows the outdoor unit in the first embodiment.

(3) FIG. 3 A perspective view that shows a part out of parts of the outdoor unit in the first embodiment where a gripping member is attached.

(4) FIG. 4 A perspective view that shows the gripping member in the first embodiment.

(5) FIG. 5 A view of the gripping member in the first embodiment as seen from the front side.

(6) FIG. 6 A cross-sectional view that shows the gripping member in the first embodiment, and corresponds to the cross-sectional line VI-VI in FIG. 5.

(7) FIG. 7 A cross-sectional view that shows the gripping member in the first embodiment, and corresponds to the cross-sectional line VII-VII in FIG. 5.

(8) FIG. 8 A view of the gripping members in the first embodiment as seen from the top side.

(9) FIG. 9 A cross-sectional view that shows a portion of attachment procedures of the gripping member to a housing in the first embodiment.

(10) FIG. 10 A cross-sectional view that shows another portion of the installation procedures of the gripping member to the housing in the first embodiment.

(11) FIG. 11 A perspective view that shows the gripping members in a second embodiment.

(12) FIG. 12 A view of the gripping members in the second embodiment as seen from the top side.

(13) FIG. 13 A view of the gripping members in a third embodiment as seen from the top side.

(14) FIG. 14 A view of the gripping members in a fourth embodiment as seen from the top side.

(15) FIG. 15 A view of the gripping members in a fifth embodiment as seen from the top side.

DETAILED DESCRIPTION

(16) Hereinafter, an embodiment is explained with reference to drawings. The scope of the present disclosure is not limited to the embodiments below, and may be changed appropriately so long as the changes are within the scope of the technical idea of the present disclosure. In order to facilitate better understanding of the various configurations, there are cases where scales and numbers or the like of various configurations in the drawings below differ from scales and numbers of actual configurations.

(17) An X axis, a Y axis, and a Z axis are shown in the drawings as deemed appropriately. The X axis shows a front-back direction of an outdoor unit **10** in the embodiments below. The Y axis shows a width direction of the outdoor unit that is orthogonal to the front-back direction. The Z axis shows a vertical direction. The front-back direction, the width direction, and the vertical direction are directions that mutually cross one another. Hereinafter, the front-back direction is referred to as a “front-back direction X”, the width direction is referred to as a “width direction Y”, and the vertical direction is referred to as a “vertical direction Z”. The vertical direction, or a Z axis direction corresponds to a “predetermined direction”.

(18) Out of sides of the front-back direction X, a side to which an arrow of the X axis faces (+X side) is a front side of the outdoor unit. Out of the sides of the front-back direction X, an opposite side to which the arrow of the X axis faces (−X side) is a back side of the outdoor unit. The width direction Y is a left-right direction of the outdoor unit. Out of the sides of the width direction Y, a side to which the arrow of the Y axis faces (+Y side) is a left side of the outdoor unit. Out of the sides of the width direction Y, an opposite side to which the arrow of the Y axis faces (−Y side) is a right side of the outdoor unit. In the description below, the side out of the sides of the width direction Y to which the arrow of the Y axis faces (+Y side), in other words the left side, is referred to as a “one side of the width direction”. The opposite side out of the sides of the width direction Y to which the arrow of the Y axis faces (−Y side) is referred to as an “other side of the width direction”. A side out of the sides of the vertical direction Z to which the arrow of the Z axis faces (+Z side) is a top side. An opposite side out of the sides of the vertical direction Z to which the arrow of the Z axis faces (−Z side) is a bottom side.

First Embodiment

(19) FIG. 1 is a schematic block diagram that schematically shows a refrigeration cycle device **100** which is included in the outdoor unit **10** in a first embodiment. The refrigeration cycle device **100** is a device which utilizes a refrigeration cycle that circulates a refrigerant **40**. The refrigeration cycle device **100** in the first embodiment is an air conditioning device. As shown in FIG. 1, the refrigeration cycle device **100** includes the outdoor unit **10**, an indoor unit **20**, and a circulation path **30**. The outdoor unit **10** is disposed outdoors. The indoor unit **20** is disposed indoors. The outdoor unit **10** and the indoor unit **20** are mutually connected by the circulation path **30** that

circulates the refrigerant **40**. The outdoor unit **10** and the indoor unit **20** make up a heat exchange unit that conducts heat exchange between the units thereof and the surrounding atmosphere.

(20) By the refrigeration cycle device **100** conducting heat exchange between the refrigerant **40** circulating inside the circulation path **30** and the indoor atmosphere where the indoor unit **20** is disposed, the temperature of the indoor atmosphere is adjustable. A refrigerant such as a fluorine based refrigerant with a low global warming potential (GWP: Global Warming Potential), or a hydrocarbon based refrigerant or the like may be mentioned as examples of the refrigerant **40**.

(21) The outdoor unit **10** has a housing **11**, a compressor **12**, a heat exchange device **13**, a flow adjustment valve **14**, a blower **15**, a four-way valve **16**, and a controller **17**. The compressor **12**, the heat exchange device **13**, the flow adjustment valve **14**, the blower **15**, the four-way valve **16**, and the controller **17** are housed on an inside of housing **11**. The controller **17** controls the various parts of the outdoor unit **10**. The controller **17** for example, may be a system controller that regulates the control of the entirety of the refrigeration cycle device **100**.

(22) Out of components on the circulation path **30**, the compressor **12**, the heat exchange device **13**, the flow adjustment valve **14**, and the four-way valve **16** are provided on a part located on the inside of housing **11**. Out of the components on the circulation path **30**, the compressor **12**, the heat exchange device **13**, the flow adjustment valve **14**, and the four-way valve **16** are connected by the part located on the inside of housing **11**.

(23) The four-way valve **16** is provided on a portion connected to a discharge side of the compressor **12** out of components on the circulation path **30**. The four-way valve **16** is able to reverse the flow of the refrigerant **40** inside the circulation path **30** by switching over a portion of the path of the circulation path **30**. When the path connected by the four-way valve **16** is the path of the four-way valve **16** that is shown by solid lines in FIG. **1**, the refrigerant **40** flows in the direction of the circulation path **30** shown by the solid line arrow in FIG. **1**. On the other hand, when the path connected by the four-way valve **16** is the path of the four-way valve **16** that is shown by dashed lines in FIG. **1**, the refrigerant **40** flows in the direction of the circulation path **30** shown by the dashed line arrow in FIG. **1**.

(24) The indoor unit **20** has a housing **21**, a heat exchange device **22**, and a blower **23**. The heat exchange device **22** and the blower **23** are housed on an inside of housing **21**. The indoor unit **20** is able to cool the indoor atmosphere the indoor unit **20** is disposed in, through a cooling operation. The indoor unit **20** is able to warm the indoor atmosphere the indoor unit **20** is disposed in, through a heating operation.

(25) When the indoor unit **20** is in the cooling operation, the refrigerant **40** that flows within the circulation path **30** flows in the direction of the solid line arrow shown in FIG. **1**. In other words, when the indoor unit **20** is in cooling operation, the refrigerant **40** that flows within the circulation path **30** circulates so as to return to the compressor **12** after passing through the compressor **12**, the heat exchange device **13** of the outdoor unit **10**, the flow adjustment valve **14**, and the heat exchange device **22** of the indoor unit **20** in such an order. During the cooling operation, the heat exchange device **13** inside the outdoor unit **10** functions as a condenser, and the heat exchange device **22** of the indoor unit **20** functions as an evaporator.

(26) On the other hand, when the indoor unit **20** is in the heating operation, the refrigerant **40** that flows within the circulation path **30** flows in the direction of the dashed line arrow shown in FIG. **1**. In other words, when the indoor unit **20** is in heating operation, the refrigerant **40** that flows within the circulation path **30** circulates so as to return to the compressor **12** after passing through the compressor **12**, the heat exchange device **22** of the indoor unit **20**, the flow adjustment valve **14**, and the heat exchange device **13** of the outdoor unit **10** in such an order. During the heating operation, the heat exchange device **13** inside the outdoor unit **10** functions as an evaporator, and the heat exchange device **22** of the indoor unit **20** functions as a condenser.

(27) Next, the outdoor unit **10** in the first embodiment is further explained. FIG. **2** is a perspective view that shows the outdoor unit **10**. As shown in FIG. **2**, the housing **11** is roughly a rectangular

box-shape having surfaces that face each of the front-back direction X, the width direction Y, and the vertical direction Z. In the first embodiment, the housing **11** has two blowers **15** that are provided so as to be aligned in the vertical direction Z. The housing **11** has a front surface panel **11a** that is attached so as to be detachable. The front surface panel **11a** is a sheet metal member. The front surface panel **11a** has a front wall portion **11b** that configures a portion of a front side (+X side) wall portion in the housing **11**, and a side wall portion **11c** that configures a portion of the one side of the width direction (+Y side) wall portion in the housing **11**. The side wall portion **11c** bends to the back side (-X side), from the one side of the width direction of the front wall portion **11b**.

(28) The outdoor unit **10** includes gripping members **50**. The gripping members **50** are members to which an operator or the like may hook their hands or fingers to when transporting the outdoor unit **10**. The gripping members **50** in the first embodiment are resin members that are integrally formed by a metal mold of injection molding or the like. The resin that the gripping members **50** are configured from may for example be polypropylene or the like. A plurality of the gripping members **50** are provided. The gripping members **50** in the first embodiment are provided on each of four corner portions out of portions of the housing **11** to which the front-back direction X wall portion and the width direction Y wall portion are connected to. The gripping members **50** in the first embodiment are located on the bottom side in the vertical direction Z more than a center of the outdoor unit **10**.

(29) FIG. **3** is a perspective view that shows a part out of parts of the outdoor unit **10** where a gripping member **50** is attached. FIG. **4** is a perspective view that shows the gripping member **50**. FIG. **5** is a view of the gripping member **50** as seen from the front side (+X side). FIG. **6** is a cross-sectional view that shows the gripping member **50**, and corresponds to the cross-sectional line VI-VI in FIG. **5**. FIG. **7** is a cross-sectional view that shows the gripping member **50**, and corresponds to the cross-sectional line VII-VII in FIG. **5**. FIG. **8** is a view of the gripping member **50** as seen from the top side.

(30) The gripping members **50** shown in FIG. **3** to FIG. **8** are gripping members **50** that are provided in the corner portions out of portions of the housing **11** to which the front side (+X side) wall portion and the one side of the width direction (+Y side) wall portion are connected to. The corner portions, to which the gripping members **50** are attached, shown in FIG. **3** to FIG. **8** are provided on the front surface panel **11a**. Although the four gripping member **50** are each attached in differing locations and orientations, each have the same construction. For such reason, the relative positional relationship and the like of the gripping member **50** attached to the front surface panel **11a**, as a representative of the four gripping members **50**, is described in the explanation of the gripping member **50** below.

(31) As shown in FIG. **3**, the gripping member **50** is attached to an opening **11d** that is provided in the housing **11**. The opening **11d** penetrates the wall portion that configures the housing **11**. The opening **11d** shown in FIG. **3** is provided so as to extend over the front wall portion **11b** and the side wall portion **11c** in the front surface panel **11a**. An opening direction of the opening **11d** in the first embodiment is a direction that is orthogonal to the vertical direction Z. In other words, the vertical direction Z is orthogonal to the opening direction of the opening lid. The opening lid shown in FIG. **3** opens to the front side (+X side) and to the one side of the width direction (+Y side). The opening **11d** is covered by attaching the gripping member **50**. A protrusion **11e** that protrudes to the inside of housing **11** is provided on an inner edge of the opening **11d**, as shown in FIG. **6** and FIG. **7**. The protrusion **11e** is at least provided on the portions out of portions of the inner edge of the opening **11d** that face a plurality of claws **60**, to be mentioned later on.

(32) As shown in FIG. **4** to FIG. **8**, the gripping member **50** has a main body **51** and a frame body **52**. The main body **51** is roughly a rectangular box-shape that opens to the front side (+X side) and to the one side of the width direction (+Y side). As shown in FIG. **6** and FIG. **7**, the main body **51** is housed on the inside of housing **11**.

(33) The main body **51** has an upper wall portion **51a**, a bottom wall portion **51b**, an inner wall portion **51c**, and a side wall portion **51d**. Out of wall portions of the main body **51**, the upper wall portion **51a** is the wall portion that is located on the top side. Out of wall portions of the main body **51**, the bottom wall portion **51b** is the wall portion that is located on the bottom side. Out of wall portions of the main body **51**, the inner wall portion **51c** is the wall portion located on the back side ($-X$ side). Out of wall portions of the main body **51**, the side wall portion **51d** is the wall portion located on the other side of the width direction ($-Y$ side).

(34) As shown in FIG. 4, the upper wall portion **51a** has through holes **54** that penetrate the upper wall portion **51a** in the vertical direction Z . A pair of through holes **54** are provided at an interval in the width direction Y in the first embodiment. The through holes **54** are provided on the front side ($+X$ side) portion of the upper wall portion **51a**. As shown in FIG. 8, the through holes **54** are roughly long rectangular shapes in the width direction Y , as seen from the vertical direction Z . Edges of the front side of the through holes **54** are disposed so as to be spaced apart in the back side ($-X$ side) more than the edge of the front side of the upper wall portion **51a**.

(35) As shown in FIG. 6 and FIG. 7, the frame body **52** is connected to the opening edge **51e** of the main body **51**. The frame body **52** is a frame shape along the opening edge **51e** of the main body **51**. The frame body **52** protrudes from the opening edge **51e** of the main body **51** to the outside over the entire circumference of the opening edge **51e**. The frame body **52** is located on the outside of housing **11**. Out of parts of the frame body **52**, the parts that protrude to the outside more than the main body **51** are coming into contact with, and hook onto, a circumferential edge of the opening **11d** from an outside of housing **11**, out of the outside surfaces of the housing **11**. The circumferential edge of the opening **11d** is sandwiched between the plurality of claws **60** to be mentioned later on, and the frame body **52**.

(36) As shown in FIG. 4, the frame body **52** has a first portion **52a** that is connected to front side ($+X$ side) of the main body **51**, and a second portion **52b** that is connected to the one side of the width direction ($+Y$ side) of the main body **51**. The first portion **52a** protrudes to both sides of the vertical direction Z and to the other side of the width direction ($-Y$ side) more than the main body **51**. The second portion **52b** extends to the back side ($-X$ side) from an end portion of the one side of the width direction of the first portion **52a**. The second portion **52b** protrudes to both sides of the vertical direction Z and the back side more than the main body **51**.

(37) The inside of frame body **52** is an insertion opening **52c** that is connected to the inside of main body **51**. The frame body **52** has a cover portion **52d** that covers atop side portion out of portions of the opening of the main body **51**. A hand insertion space **53** (inside space) is configured by the main body **51** and the frame body **52**. The hand insertion space **53** is a space into which the operator is able to hook their hands or fingers to when transporting the outdoor unit **10**. The hand insertion space **53** opens to the outside of housing **11** via the insertion opening **52c**. The hand insertion space **53** is a space that is provided on the inside of the main body **51**. The operator transporting the outdoor unit **10** inserts their hands or fingers in the hand insertion space **53** and holds the cover portion **52d**. Therefore, it is possible for the operator to lift the outdoor unit **10** by holding onto the gripping member **50**.

(38) As shown in FIG. 6, the gripping member **50** has the plurality of claws **60**. The plurality of claws **60** protrude from the main body **51**. The plurality of claws **60** are located on the inside of housing **11**. The plurality of claws **60** are disposed so as to face the circumferential edge of the openings **11d** in the inside of housing **11**. The plurality of claws **60** are disposed so as to face an inner end portion of the protrusion **11e** that is provided in the opening **11d**. The plurality of claws **60** hook onto the circumferential edge of the openings **11d** from the inside of the housing **11**.

(39) As shown in FIG. 5, the plurality of the claws **60** include first claw portions **60a**, and second claw portions **60b**. The first claw portions **60a** are claws **60** that are provided on the top side of the main body **51**. The second claw portions **60b** are claws **60** that are provided on the bottom side of the main body **51**. In other words, the second claw portions **60b** are provided on the opposite side

of the first claw portions **60a** in the vertical direction Z. Each of a plurality of first claw portions **60a** and of second claw portions **60b** are provided in the first embodiment. The plurality of first claw portions **60a** protrude to the top side from the upper wall portion **51a** of the main body **51**. The plurality of the second claw portions **60b** protrude to the bottom side from the bottom wall portion **51b** of the main body **51**.

(40) The first claw portions **60a** include a fixed claw **61**, and elastic claws **62**. The fixed claw **61** and the elastic claws **62** protrude to the top side from the upper wall portion **51a** of the main body **51**. As shown in FIG. 8, the fixed claw **61** in the first embodiment is provided between a pair of through holes **54** on a portion out of portions of the upper wall portion **51a** in the width direction Y. The fixed claw **61** is located on a center portion of the upper wall portion **51a** in the width direction Y. A bottom end portion of the fixed claw **61** is entirely connected to the upper wall portion **51a**. As shown in FIG. 4 and FIG. 6, the fixed claw **61** is roughly a triangular shape, as seen from the width direction Y. The protrusion height of the fixed claw **61** with respect to the main body **51** becomes larger as the fixed claw **61** approaches the front side (+X side) of the main body **51**. The protrusion height of the fixed claw **61** with respect to the main body **51** is a vertical direction Z dimension of the fixed claw **61**, and represents a distance between a top surface of the upper wall portion **51a** and a top end portion of the fixed claw **61** in the vertical direction Z. The top end portion of the fixed claw **61** in the first embodiment is located more to the bottom side than the top end portion of the frame body **52**.

(41) As shown in FIG. 6, the fixed claw **61** has a first facing surface **61f** that faces the circumferential edge of the openings **11d**, and a first inclined surface **61i**. The first facing surface **61f** faces the front side (+X side). The first facing surface **61f** is a surface that is roughly parallel with the vertical direction Z. For example, although not shown in the figures, the first facing surface **61f** is a surface that slightly inclines towards a direction located in the back side with respect to the vertical direction Z as the first facing surface **61f** approaches the top side. The first facing surface **61f** faces the inner end portion of protrusion **11e** in the first embodiment. The first facing surface **61f** contacts the inner end portion of protrusion **11e**. The first inclined surface **61i** is a surface that faces the back side, and faces an inclined top side. A top end portion of the first inclined surface **61i** is connected to a top end portion of the first facing surface **61f**. The first inclined surface **61i** becomes located on the top side as the first inclined surface **61i** moves towards the front side.

(42) When the first facing surface **61f** is hooked onto the protrusion **11e**, the fixed claw **61** is hooked from the inside of housing **11** to the circumferential edge of the openings **11d**. A portion out of portions of the circumferential edge of the openings **11d** that is located on the top side is sandwiched in the front-back direction X by a portion that protrudes to the top side more than the main body **51** out of portions of the frame body **52**, and by the fixed claw **61**.

(43) As shown in FIG. 4, the elastic claws **62** in the first embodiment are provided in a pair that sandwiches the fixed claw **61** in the width direction Y on the upper wall portion **51a**. Each elastic claw **62** in the pair of elastic claws **62** has the same shape. The pair of elastic claws **62** are each disposed inside the pair of through holes **54**. The pair of elastic claws **62** are each cantilevered by an inner edge of each of the pair of through holes **54**. An end of each of the elastic claw **62** in the back side (-X side) is connected to a portion out of portions of the through holes **54** that is located to the back side of the inner edge of each of the through holes **54**. In other words, the elastic claws **62** in the first embodiment are connected from the opening **11d** out of the inner edges of through hole **54**, to an edge of a far side of through hole **54**. Therefore, each of the elastic claws **62** is cantilevered by the main body **51**. A gap is provided on the edges of both sides of the width direction between the elastic claws **62** and the through holes **54**, and on the edge of the front side between the elastic claws **62** and the through holes **54**.

(44) As shown in FIG. 7, an elastic claw **62** of the elastic claws **62** has a support portion **62a** and a claw main body **62b**. The support portions **62a** in the first embodiment are plate shapes that have

plate surfaces which face the vertical direction Z. The support portions **62a** are located inside the through holes **54**. The support portions **62a** extend from the portion out of portions of inner edges of each of the through holes **54** that is located on the back side ($-X$ side) to the front side ($+X$ side). An end of the support portions **62a** front side is disposed so as to face the back side of the portion that is located on the front side of the through holes **54** out of portions of the inner edges of the through holes **54**, with a gap in between.

(45) The claw main bodies **62b** protrude to the top side from the portion in the front side ($+X$ side) of the support portions **62a**. The claw main bodies **62b** protrude to the top side more than the through holes **54**. The claw main bodies **62b** are roughly triangular shapes, as seen from the width direction Y. Protrusion heights of the claw main bodies **62b** with respect to the main body **51** become larger as the claw main bodies **62b** approach the front side of the main body **51**. The protrusion heights of the claw main bodies **62b** with respect to the main body **51** are vertical direction Z dimensions of the claw main bodies **62b**, and are protrusion heights of the elastic claws **62** with respect to the main body **51**. The protrusion heights of the elastic claws **62** with respect to the main body **51** represent distances between the top surface of the upper wall portion **51a** and top end portions of the elastic claws **62** in the vertical direction Z.

(46) As shown in FIG. 5, top end portions of the elastic claws **62** in the first embodiment, in other words the top end portions of the claw main bodies **62b**, are located on the top side more than the top end portion of the fixed claw **61**. In other words, the protrusion height of the fixed claw **61** with respect to the main body **51** is less than the protrusion heights of the elastic claws **62** with respect to the main body **51**. The top end portions of the elastic claws **62** are located in the bottom side more than the top end portion of the frame body **52**.

(47) As shown in FIG. 7, the elastic claws **62** have second facing surfaces **62f** that face the circumferential edge of the openings **11d**, and second inclined surfaces **62i**. The second facing surfaces **62f** face the front side ($+X$ side). The second facing surfaces **62f** are surfaces that are roughly parallel with the vertical direction Z. For example, although not shown in the figures, the second facing surfaces **62f** are surfaces that slightly incline towards a direction located in the back side with respect to the vertical direction Z as the second facing surfaces **62f** approach the top side. The second facing surfaces **62f** are connected to the top side of the support portions **62a** front side surface. The second facing surfaces **62f** in the first embodiment face inner end portion of the protrusion **11e**. The second facing surfaces **62f** are in contact with the inner end portion of the protrusion **11e**. The second inclined surfaces **62i** are surfaces that face the back side ($-X$ side), and incline towards the top side. The second inclined surfaces **62i** top sides are connected to the second facing surfaces **62f** top sides. The second inclined surfaces **62i** become located on the top side as the second inclined surfaces **62i** move towards the front side.

(48) When the second facing surfaces **62f** are hooked to the protrusion **11e**, the elastic claws **62** are hooked from the inside of the housing **11** to the circumferential edge of the openings **11d**. A portion out of portions of the circumferential edge of the openings **11d** that is located on the top side is sandwiched in the front-back direction X by a portion that protrudes to the top side more than the main body **51** out of portions of the frame body **52**, and by the elastic claws **62**.

(49) As shown in FIG. 8, a dimension of the elastic claws **62** in the width direction Y is greater than a dimension of the fixed claw **61** in the width direction Y. Surfaces of the elastic claws **62** front sides ($+X$ side) are located slightly more to the front side than a surface of the fixed claw **61** front side. In other words, the second facing surfaces **62f** are located on a side closer to the circumferential edge of the openings **11d** than the first facing surface **61f**.

(50) The elastic claws **62** deform elastically in the vertical direction Z with respect to the main body **51**. Specifically, by making connection portions of the inner edges of the through holes **54** in the support portions **62a** elastically deformable in the vertical direction Z, it is possible to have the elastic claws **62** elastically deform in the vertical direction Z with the connection portions of the support portions **62a** and the inner edges of the through holes **54** as a pivot.

(51) As shown in FIG. 5, the second claw portions **60b** include fixed claws **63**. The fixed claws **63** protrude to the bottom side from the bottom wall portion **51b** of the main body **51**. A pair of fixed claws **63** in the first embodiment is provided in the width direction Y with a space in between. The pair of fixed claws **63** are each disposed in a location that overlaps with the pair of elastic claws **62** as seen from the vertical direction Z. Top end portions of the fixed claws **63** are entirely connected to the bottom wall portion **51b**. As shown in FIG. 7, the fixed claws **63** are roughly shaped as isosceles triangles, as seen from the width direction Y. Protrusion heights of the fixed claws **63** with respect to the main body **51**, in other words the protrusion heights of the second claw portions **60b** with respect to the main body **51**, are greater than the protrusion heights of the elastic claws **62** with respect to the main body **51**. The protrusion heights of the fixed claws **63** with respect to the main body **51** in the first embodiment are greater than the protrusion heights of elastic claws **62** with respect to the main body **51**. The protrusion heights of the fixed claws **63** with respect to the main body **51** are vertical direction Z dimensions of the fixed claws **63**, and represent distances between a bottom surface of the bottom wall portion **51b** and a bottom end portions of fixed claws **63** in the vertical direction Z. The bottom end portions of fixed claws **63** in the first embodiment are located more to the bottom side than a bottom end portion of frame body **52**.

(52) The fixed claws **63** have third facing surfaces **63f** that face the circumferential edge of the openings **11d**. The third facing surfaces **63f** face the front side (+X side). The third facing surfaces **63f** are surfaces that incline towards a direction located in the back side (-X side) with respect to the vertical direction Z as the third facing surfaces **63f** approach the bottom side. The third facing surfaces **63f** face the inner end portion of the protrusion **11e** in the first embodiment. The third facing surfaces **63f** contact the inner end portion of the protrusion **11e**. A surface on the back side of the fixed claws **63** is symmetrically disposed to the third facing surfaces **63f** in the front back direction X.

(53) When the third facing surfaces **63f** are hooked onto the protrusion **11e**, the third facing surfaces **63f** are hooked from the inside of the housing **11** to the circumferential edge of the openings **11d**. A portion out of portions of the circumferential edge of the openings **11d** that is located on the bottom side is sandwiched in the front-back direction X by a portion that protrudes to the bottom side more than the main body **51** out of portions of the frame body **52**, and by the fixed claws **63**.

(54) FIG. 9 is a cross-sectional view that shows a portion of attachment procedures of the gripping member **50** to the housing **11**. FIG. 10 is a cross-sectional view that shows another portion of the attachment procedures of the gripping member **50** to the housing **11** in the first embodiment. The operator attaching the gripping members **50** to the housing **11** brings the gripping members **50** close to the openings **11d** from an outside of the housing **11**. The operator then tilts the gripping member **50** with respect to the vertical direction Z, and inserts the fixed claws **63** provided on a bottom side portion of the main body **51** and on the bottom side of the main body **51** to an inside of the openings **11d**.

(55) In a state where the fixed claws **63** are hooked to circumferential edge of the openings **11d** on the inside of the housing **11**, the operator inserts the top side portion of the main body **51** as shown in FIG. 9 to the inside of the openings **11d**, straightening out the tilt of the gripping members **50** with respect to the vertical Z direction. In FIG. 9 and in FIG. 10, although a state where the top side portion of the main body **51** is moved straight in the back side (-X side) and is inserted to the inside of the opening **11d**, it is possible to insert the top side portion of the main body **51** to the inside of the opening **11d** by rotating the gripping members **50**, with the bottom side portion of the gripping members **50** that are hooked onto the opening **11d** as a pivot.

(56) After inserting the top side portion of the main body **51** to the inside of the opening **11d**, the second inclined surfaces **62i** of the elastic claws **62** contact the circumferential edge of the top side of the opening **11d**. In this state, when the gripping members **50** are further pressed into the openings **11d**, as shown in FIG. 10, the second inclined surfaces **62i** are pressed by the

circumferential edge of the openings **11d**, and the elastic claws **62** deform in the direction ($-Z$ direction) in which the elastic claws **62** are push-inserted to the inside of the main body **51**, where the connection portions of the support portions **62a** and the inner edges of the through holes **54** act as pivots. Therefore, it is possible to decrease the protrusion heights of elastic claws **62** with respect to the main body **51**, and to easily insert the elastic claws **62** to the inside of the openings **11d**.

(57) When the main body **51** is further pressed to the inside of opening **11d**, the elastic claws **62** entirely pass through the opening **11d**, causing the elastically deformed support portions **62a** to restore, and the elastic claws **62** return to the original state of the elastic claws **62**. Therefore, the second facing surfaces **62f** of the elastic claws **62** are in a state where the second facing surfaces **62f** of the elastic claws **62** face the inner end portion of the protrusion **11e**, and the elastic claws **62** are hooked onto the circumferential edge of the openings **11d** from the inside of the housing **11**.

(58) As previously mentioned, when the gripping members **50** are push-inserted to the inside of the openings **11d**, as with elastic claws **62**, the fixed claw **61** provided on the top side of the main body **51** also contacts the edge of the top side of the openings **11d**. In such case, the protrusion height of the fixed claw **61** is less than the protrusion heights of elastic claws **62**. As such, when push-inserting a gripping member **50** of the gripping members **50** to the inside of opening **11d**, by having at least one of the inner edge of the opening **11d** or the fixed claw **61** slightly deform elastically, the fixed claw **61** is inserted into the housing **11** through the opening lid. After the fixed claw **61** passes through the opening lid, the elastically deformed portion is restored, and the first facing surface **61f** of the fixed claw **61** faces the inner end portion of protrusion **11e**. Therefore, the fixed claw **61** is hooked onto the circumferential edge of the openings **11d** from the inside of the housing **11**.

(59) As shown above, it is possible to attach the gripping members **50** to the openings **11d** of the housing **11** by push-inserting the gripping members **50** to the inside of the openings **11d** from the outside of the housing **11**. In a state where the gripping members **50** are attached to the housing **11**, the circumferential edge of the openings **11d** is sandwiched by the plurality of claws **60** and by the frame body **52**. Therefore, it is possible to suppress the gripping members **50** from coming off of the opening **11d**.

(60) In a case where conventional gripping members are inserted and attached to the openings **11d** from the outside of housing **11**, as with the gripping members **50** mentioned above, after inserting the claws provided on one side in the vertical direction Z to the inside of the openings **11d**, all of the claws that are provided on the other side in the vertical direction Z are pressed into the inside of the openings **11d**, as with fixed claws **61** previously mentioned. In other words, at least one of the inner edges of the openings **11d** and all of the claws provided in the other side in the vertical direction Z is slightly deformed elastically when push-inserting the above claws to the inside of openings **11d**. In such case, when the height of protrusions of the elastically deformable claws that are push-inserted are large, it is not possible to push and insert the claws with the claws being hooked onto the inner edges of the openings **11d** to the inside of the openings **11d**, and there is a need to decrease the size of the protrusion heights of the claws. As such, when push-inserting and attaching the conventional gripping members to the inside of openings **11d** from the outside of housing **11**, it is difficult for the claws to have enough protrusion heights, and so attaching the gripping members firmly with respect to the housing **11** is difficult.

(61) In contrast to the above, the elastic claws **62** according to the first embodiment are provided so as to be cantilevered by the main body **51**. The elastic claws **62** are elastically deformable in the vertical direction Z with respect to the main body **51**. As such, by elastically deforming the elastic claws **62** as previously mentioned, it is possible to insert the gripping members **50** into the openings **11d** in a state where the protrusion heights of the elastic claws **62** with respect to the main body **51** are smaller. After the elastic claws **62** pass through the openings **11d**, the elastic claws **62** restore after deforming, and the protrusion heights of elastic claws **62** with respect to the main body **51** return to the protrusion heights before deforming. As such, even if the protrusion heights of elastic claws **62** are made to be large to some extent, by push-inserting the gripping members **50**

into the openings **11d**, it is possible to easily insert the elastic claws **62** into the housing **11**. Therefore, it is possible to make each of protrusion heights of claws **60** provided on both sides in the vertical direction Z of the main body **51** larger, and it is possible to firmly attach the gripping members **50** to the openings **11d**.

(62) From the above, according to the first embodiment, it is possible to ensure an attachment strength of the gripping members **50**, along with improving work efficiency of the gripping members **50**. When attaching the gripping members **50**, it is possible for the operator to easily confirm the attachment of the gripping members **50** to the openings **11d** by sounds or feeling and so on that result when the elastic claws **62** restore after deforming. As such, it is possible to increase the work efficiency of the gripping members **50**.

(63) By push-inserting the gripping members **50** into the openings **11d** from the outside of the housing **11**, by providing the openings **11d** in the housing **11** in order to attach the gripping members **50** to the housing **11**, it is possible to attach the gripping members **50** to any location of the housing **11**. As such, compared to a case where the gripping members **50** are held by being sandwiched between two or more members that configure the housing **11**, it is possible to improve a degree of freedom of design of attachment locations of the gripping members **50** in the housing **11**. Therefore, it is possible to attach the gripping members **50** to locations where it is easier to hold the gripping members **50**, out of locations of the housing **11**.

(64) The elastic claws **62** according to the first embodiment are the first claw portions **60a**. The protrusion heights of the second claw portions **60b** with respect to the main body **51** are greater than the protrusion heights of the elastic claws **62** with respect to the main body **51**. As such, it is possible to increase the protrusion heights of the second claw portions **60b** to be greater than or equal to the protrusion heights of the elastic claws **62**, protrusion heights of that can be relatively great, as previously mentioned. As such, it is possible to appropriately increase each of the protrusion heights of the elastic claws **62**, which are the first claw portions **60a**, and of the second claw portions **60b**. Therefore, it is possible to firmly attach the gripping members **50** to the openings **11d**.

(65) According to the first embodiment, the main body **51** has through holes **54** that are configured to penetrate the wall portion of the main body **51**. The elastic claws **62** are cantilevered by the inner edges of the through holes **54**. As such, it is possible to easily have the elastic claws **62** be cantilevered by the main body **51**. It is also possible to easily have the elastic claws **62** elastically deform via the through holes **54**.

(66) According to the first embodiment, the elastic claws **62** in the first embodiment are connected to edges, out of the inner edges of through hole **54**, of the sides that are far from the opening **11d**. As such, compared to a case where the elastic claws **62** are connected to the edges closer to the opening **11d** out of the inner edges of through hole **54**, it is easier to have the elastic claws **62** be in a separable orientation from the inner edges of openings **11d** when push-inserting the gripping members **50** into the openings lid. Therefore, it is possible to elastically deform the elastic claws **62**, and it is possible to easily attach the gripping members **50** to the openings **11d**.

(67) According to the first embodiment, the elastic claws **62** are provided on the top side of the main body **51** in the vertical direction. As such, the through holes **54** are also provided on the upper wall portion **51a** of the main body **51**. Therefore, compared to a case where the through holes **54** are provided on the bottom wall portion **51b** of the main body **51**, it is possible to suppress foreign bodies such as rain water or the like from entering through the through holes **54** to the housing **11**.

(68) According to the first embodiment, the plurality of first claw portions **60a** are provided. The plurality of first claw portions **60a** include the fixed claws **61** and the elastic claws **62**. As such, along with the elastic claws **62**, it is possible to also hook the fixed claws **61** to the circumferential edge of the opening **11d**. As such, it is possible to attach the gripping members **50** to the openings **11d** more firmly. The protrusion heights of the fixed claws **61** with respect to the main body **51** are less than the protrusion heights of the elastic claws **62** with respect to the main body **51**. Therefore,

even if the fixed claws **61** are not configured so as to be cantilevered, such as the case of the elastic claws **62**, it is possible to push-insert the fixed claws **61** into the opening **11d**. By providing the fixed claws **61** so as not to be cantilevered, a force applied to the gripping members **50** when the outdoor unit **10** is transported or the like is not only absorbed by the elastic claws **62**, but by the fixed claws **61** as well. As such, it is possible to prevent excess load from being applied to the elastic claws **62**. Therefore, it is possible to prevent concentration of stresses on portions out of the elastic claws **62** that are connected to the through holes **54**, in other words a base portion of the support portions **62a**, and it is possible to prevent the elastic claws **62** from being damaged.

(69) According to the first embodiment, the second facing surfaces **62f** of the elastic claws **62** are located on a side closer to the circumferential edge of the openings **11d** than the first facing surfaces **61f** of the fixed claws **61**. As such, in a case where the gripping members **50** are pulled to the outside of the housing **11** during transport or the like of the outdoor unit **10**, it is possible to have the second facing surfaces **62f** of the elastic claws **62** to come into contact with the circumferential edge of openings **11d** before the first facing surface **61f** of the fixed claws **61**. As such, it is possible to prevent the fixed claws **61** from coming out from the openings **11d**. Therefore, it is possible to attach the gripping members **50** more securely to the openings **11d** than the conventional gripping members. When attaching the gripping members **50** to the openings **11d**, the fixed claws **61** pass through the openings **11d** before the elastic claws **62** pass through the openings **11d**. As such, after passing the elastic claws **62** through the openings **11d** and confirming that the elastic claws **62** have restored after deforming through sounds or feeling or the like, the fixed claws **61** are in a state where the fixed claws **61** have passed through the openings **11d**. Therefore, by being able to confirm that the elastic claws **62** have passed through the openings **11d** through sounds or feeling or the like, it is easy to confirm appropriate attachment of the gripping members **50** with respect to the housing **11**.

(70) According to the first embodiment, the plurality of first claw portions **60a** include a plurality of elastic claws **62**. As such, it is possible to firmly attach the gripping members **50** to the openings **11d** by the plurality of elastic claws **62**. Compared to a case where a dimension of one of the elastic claws **62** in the width direction Y is made larger so as to improve the attachment strength of the gripping members **50**, it is possible to make the dimension of each of the elastic claws **62** in the width direction Y smaller, making it easy to elastically deform each of the elastic claws **62**. Therefore, it is possible to easily attach the gripping members **50** to the openings lid.

(71) According to the first embodiment, the gripping members **50** have the frame body **52** along the frame of the opening edge **51e** of the main body **51**. The frame body **52** is located on the outside of the housing **11**, and sandwiches the circumferential edge of the openings **11d** in between with the plurality of claws **60**. As such, it is possible to stably sandwich the circumferential edge of the openings **11d**, and to more firmly attach the gripping members **50** to the openings **11d** by the frame body **52** and the plurality of claws **60**.

(72) According to the first embodiment, the heat exchange unit that includes the gripping members **50** provided with the elastic claws **62** is the outdoor unit **10**. The mass of the outdoor unit **10** is greater, compared to the indoor unit **20**. As such, the outdoor unit **10** is harder to transport, compared to the indoor unit **20**. With respect to the above, by providing the gripping members **50** that ensures the previously mentioned attachment strength, it is possible to easily transport the outdoor unit **10** via the gripping members **50**.

(73) According to the first embodiment, top end portions of first claw portions **60a** are located in the bottom side more than top end portions of frame body **52**. Bottom end portions of the second claw portions **60b** are located more to the bottom side than the bottom end portions of frame body **52**. As such, the first claw portions **60a** are hidden and are not visible by the frame body **52**, as seen from the front side (+X side) of the gripping members **50**. On the other hand, the bottom end portions of the second claw portions **60b** protrude to the bottom side more than the frame body **52**. As such, the bottom end portions of the second claw portions **60b** are visible from the front side

before attaching the gripping members **50** to the housing **11**. Therefore, the operator attaching the gripping members **50** is able to easily grasp an orientation of the gripping members **50** when attaching the gripping members **50** to the housing **11** since the second claw portions **60b** that protrude more than the frame body **52** are visible.

(74) According to the first embodiment, a surface out of surfaces of the elastic claws **62** that is located on an opposite side to the surface where the opening **11d** is located on, is the second inclined surfaces **62i** which is inclined with respect to the vertical direction Z. As such, when push-inserting the gripping members **50** into the openings lid, the circumferential edge of the openings **11d** contact the second inclined surfaces **62i**. Therefore, when push-inserting the gripping members **50** into the openings **11d**, it is difficult for the elastic claws **62** to get hooked onto the circumferential edge of the openings **11d**, and it becomes easier to elastically deform the elastic claws **62**. It is also difficult to apply an excessive force to the circumferential edge of the openings **11d**. Therefore, it is possible to prevent deformation of the openings **11d**.

Second Embodiment

(75) FIG. **11** is a perspective view that shows gripping members **250** in a second embodiment. FIG. **12** is a view of a gripping member **250** in the second embodiment as seen from the top side. In the explanation below, configurations similar to the configurations previously mentioned have the same reference symbols appropriated, and thus the explanations thereof are omitted.

(76) As shown in FIG. **11** and in FIG. **12**, the gripping members **250** in the second embodiment have an upper wall portion **251a** of main body **251**. In contrast to the upper wall portion **51a** of the first embodiment, the upper wall portion **251a** of the main body **251** only has one through hole **54** in a center portion of the upper wall portion **251a** in the width direction Y. Another configuration of the main body **251** is similar to another configuration of the main body **51** in the first embodiment.

(77) A plurality of first claw portions **260a** in the gripping members **250** includes two fixed claws **261** and one elastic claw **262**. The one elastic claw **262** is provided in the one through hole **54**. The two fixed claws **261** are disposed so as to sandwich the one elastic claw **262** in the width direction Y. Another configuration of the fixed claws **261** is similar to another configuration of the fixed claws **61** of the first embodiment. Another configuration of an elastic claw **262** is similar to another configuration of the elastic claws **62** of the first embodiment. Another configuration of the gripping members **250** is similar to another configuration of the gripping members **50** of the first embodiment.

(78) According to the second embodiment, the plurality of first claw portions **260a** includes the plurality of fixed claws **261** and the one elastic claw **262**. As such, it is possible to appropriately receive the stresses that are applied to the elastic claw **262** by the plurality of fixed claws **261**.

(79) Therefore, it is possible to prevent damage to the elastic claw **262**, as well as appropriately and safely attach the gripping members **250** to the openings lid. When attaching the gripping members **250**, when it is possible to confirm through sounds or feeling or the like whether the one elastic claw **262** has restored inside of housing **11** after deforming and is hooked onto the circumferential edge of the opening **11d**, it is then possible to confirm that the gripping members **250** are correctly attached to the openings **11d**. In other words, if it is possible to insert one elastic claw **262** into the housing **11** correctly, there is no need to confirm whether the other elastic claws **262** have been inserted into the housing **11** correctly or not. As such, it is possible to prevent attachment of the gripping members **250** to the housing **11** in a state where the elastic claws **262** are elastically deformed on the inside of opening **11d**. Therefore, it is possible to appropriately and easily attach the gripping members **250** to the openings **11d**.

Third Embodiment

(80) FIG. **13** is a view of gripping members **350** in a third embodiment as seen from the top side. In the explanation below, configurations similar to the configurations previously mentioned have the same reference symbols appropriated, and thus the explanations thereof are omitted.

(81) As shown in FIG. **13**, except that two through holes **54** have different dispositions on an upper

wall portion **351a**, a main body **351** in the gripping members **350** of the third embodiment has a similar configuration as the configuration of the main body **51** in the first embodiment.

(82) A plurality of first claw portions **360a** includes one fixed claw **361**, and two elastic claws **362**. The one fixed claw **361** is disposed adjacently to the two elastic claws **362** in the other side of the width direction ($-Y$ side). The two elastic claws **362** are disposed adjacently in the width direction Y . A dimension of the elastic claws **362** in the width direction Y is less than a dimension of the fixed claw **361** in the width direction Y . As such, it is possible for the elastic claws **362** to elastically deform when attaching to the gripping members **350**. Therefore, it is possible to easily attach the gripping members **350**.

(83) Another configuration of the fixed claw **361** is similar to another configuration of the fixed claw **61** of the first embodiment. Another configuration of the elastic claws **362** is similar to another configuration of the elastic claws **62** of the first embodiment. Another configuration of the gripping members **350** is similar to another configuration of the gripping members **50** of the first embodiment.

Fourth Embodiment

(84) FIG. **14** is a view of gripping members **450** in a fourth embodiment as seen from the top side. In the explanation below, configurations similar to the configurations previously mentioned have the same reference symbols appropriated, and thus the explanations thereof are omitted.

(85) As shown in FIG. **14**, a plurality of first claw portions **460a** includes only two elastic claws **62** in gripping members **450** of the fourth embodiment. Unlike the first embodiment, the plurality of the first claw portions **460a** do not include the fixed claw **61**. As such, compared to a case where the fixed claws **61** are provided, the gripping members **450** are easier to make. Another configuration of the gripping members **450** is similar to another configuration of the gripping members **50** of the first embodiment.

Fifth Embodiment

(86) FIG. **15** is a view of gripping members **550** in a fifth embodiment as seen from the top side. In the explanation below, configurations similar to the configurations previously mentioned have the same reference symbols appropriated, and thus the explanations thereof are omitted.

(87) As shown in FIG. **15**, except that the one through hole **54** is disposed in the one side of the width direction ($+Y$ side) on an upper wall portion **551a**, a main body **551** in the gripping members **550** of the fifth embodiment has a similar configuration as the configuration of the main body **251** in the second embodiment. A plurality of first claw portions **560a** each only include one fixed claw **561** and one elastic claw **562**. As such, by decreasing the number of first claw portions **560a** and making the gripping members **550** easy to make, the attachment strength of the gripping members **550** is improved by the fixed claw **561** and the elastic claw **562**. The one fixed claw **561** is disposed in the other side of the width direction ($-Y$ side) of the one elastic claw **562**.

(88) Another configuration of the fixed claw **561** is similar to another configuration of the fixed claw **61** in the first embodiment. Another configuration of the elastic claw **562** is similar to another configuration of the elastic claws **62** in the first embodiment. Another configuration of the gripping members **550** is similar to another configuration of the gripping members **50** in the first embodiment.

(89) Although embodiments of the present disclosure are described above, the disclosure is not limited to the aforementioned various configurations of the embodiments, and the configurations and methods mentioned below may be adopted.

(90) So long as the claw portions at least include one first claw portion, and at least include one second claw portion, any other claw portions may be included. So long as one claw portion out of the at least one first claw portion included and the at least one second claw portion included is an elastic claw, the other first claw portion or the other second claw portion may be any claw portion. Each of the first claw portion and the second claw portion may include elastic claws. All of the first claw portions and all of the second claw portions may be elastic claws. Shapes of various claw

portions may be any shape.

(91) Protrusion heights of the various claw portions with respect to the main body are not limited in any way. A protrusion height of the first claw portion with respect to the main body may be greater than a protrusion height of the second claw portion with respect to the main body. A protrusion height of the elastic claw with respect to the main body may be greater than a protrusion height of the second claw portion with respect to the main body. A relationship between protrusion heights of the various claw portions and the frame body are not limited in any way. So long as the elastic claws are cantilevered by the main body, the elastic claws may be connected to the main body in any manner. Predetermined directions in which the first claw portions and the second claw portions are provided on need not be vertical directions, so long as the directions are orthogonal to the opening direction of the openings.

(92) The heat exchange unit of the present disclosure that is provided with gripping members may be an indoor unit of a refrigeration cycle device. In such a case, an indoor unit for example, may be a floor-mounted indoor unit, or a ceiling-mounted indoor unit or the like. The indoor refrigeration cycle device that includes the heat exchange unit of the present disclosure is not limited to an air conditioning device, so long as the device utilizes a refrigeration cycle where a refrigerant is circulated. The refrigeration cycle device may be a heat pump water heater or the like.

(93) The various configurations and methods described in the above description of the present disclosure may be appropriately combined, as long as the configurations and methods thereof do not mutually contradict one another.

Claims

1. A heat exchange unit of a refrigeration cycle device, the heat exchange unit comprising: a housing having openings, and gripping members attached to the openings, wherein each gripping member of the gripping members has: a main body having an inner space and that opens to an outside of the housing, and a plurality of claws that protrude from the main body, and that hook to a circumferential edge of the openings from an inside of the housing, the plurality of claws include: a plurality of first claw portions, and at least one second claw portion that is provided on an opposite side to the plurality of first claw portions in a predetermined direction that intersects the opening direction of an opening, at least one claw portion out of the plurality of first claw portions and the at least one second claw portion is an elastic claw that is cantilevered by the main body, and the elastic claw is elastically deformable in the predetermined direction with respect to the main body, wherein the plurality of first claw portions include: at least one fixed claw, and at least one elastic claw, and the protrusion height of the at least one fixed claw with respect to the main body is less than the protrusion height of the at least one elastic claw with respect to the main body.

2. The heat exchange unit according to claim 1, wherein a protrusion height of the at least one second claw portion with respect to the main body is greater than a protrusion height of the at least one elastic claw of the plurality of first claw portions with respect to the main body.

3. The heat exchange unit according to claim 1, wherein the main body has a through hole that penetrates a wall portion that configures the main body, and the at least one elastic claw of the plurality of first claw portions is cantilevered by an inner edge of the through hole.

4. The heat exchange unit according to claim 3, wherein the at least one elastic claw of the plurality of first claw portions is connected to an edge, out of inner edge of the through hole, of a side that is far from the opening.

5. The heat exchange unit according to claim 3, wherein the at least one elastic claw of the plurality of first claw portions is provided on an upper side of the main body in a vertical direction.

6. The heat exchange unit according to claim 1, wherein the fixed claw has a first facing surface that faces the circumferential edge of the opening, the at least one elastic claw of the plurality of first claw portions has a second facing surface that faces the circumferential edge of the opening,

and the second facing surface is located on a side closer to the circumferential edge of the opening than the first facing surface.

7. The heat exchange unit according to claim 1, wherein the at least one elastic claw of the plurality of first claw portions comprises a plurality of elastic claws.

8. The heat exchange unit according to claim 1, wherein the at least one fixed claw of the plurality of first claw portions comprises a plurality of fixed claws, and the at least one elastic claw of the plurality of first claw portions comprises a single elastic claw.

9. The heat exchange unit according to claim 1, wherein each gripping member of the gripping members has a frame body of a frame shape along an opening edge of the main body, and the frame body is located on the outside of the housing, and sandwiches the circumferential edge of the opening in between with the plurality of claws.

10. The heat exchange unit according to claim 1, wherein the heat exchange unit is an outdoor unit.
